

Composable assurance for sovereign digital currency (CBDC): An evidence-gated qualification framework

Abstract

A central bank digital currency (CBDC) architecture must jointly preserve monetary singleness, privacy, financial integrity, legal finality, and operational resilience. This study presents an evidence-constrained composition framework for reserve-settled domestic conversion and payment-versus-payment across independently governed sovereign ledgers. A formal state model explores 2,013 states and 19,480 transitions under a declared two-key equivocation bound. A local eleven-service prototype binds bilateral compliance authorizations, unequal-currency quotes, and final prepare receipts to 5-of-7 quorum certificates, while testing restart, duplicate delivery, stale evidence, and conflicting votes. Synthetic privacy experiments show that shielding and batching reduce network-observer linkability under generator shift, although residual leakage remains. A registered person-level differential-privacy service enforces a twelve-release epsilon budget. Across sixteen synthetic AML graphs, cross-institution features improve mean PR-AUC by 0.058, with heterogeneity across graph families. Calibrated sanctions screening exposes the operational trade-off between recall and false-alert workload. Economic analysis replaces a universal holding limit with a partially identified adaptive rule. A queueing digital twin passes declared 2,000-20,000 TPS scenarios but fails an overload control. The contribution is a reproducible qualification pathway, not evidence of production or national deployment.

Keywords: central bank digital currency; cross-border settlement; differential privacy; AML/CFT; operational resilience; reproducibility

Key messages

- CBDC assurance must be evaluated as a composed system, not isolated modules.
- Cross-border decisions bind monetary, FX, compliance, and prepare evidence.
- Privacy, AML, sanctions, and policy claims remain bounded by their evidence.
- Scale qualification must progress from models to physical and field evidence.

1. Introduction

CBDC work has advanced from conceptual ledgers to two-tier prototypes, privacy mechanisms, offline payment designs, programmable compliance and cross-border experiments. The 2024 BIS survey reported broad central-bank exploration, but exploration does not imply that one architecture has jointly established monetary correctness, privacy, financial-integrity effectiveness, economic stability and national-scale resilience (Illes, Kosse, and Wierst 2025). The central challenge is compositional assurance: improving one property can weaken another. Detailed transaction visibility may assist investigation but create population-scale surveillance; aggressive privacy may weaken typology detection; immediate convertibility may increase usability while accelerating deposit flight; cross-system coordination may improve reach while creating a shared governance or failure concentration.

This paper studies the following question: under which explicitly testable assumptions can independent sovereign CBDC systems provide reserve-settled conversion, privacy-preserving financial integrity, evidence-bound cross-border finality, adaptive policy limits and a falsifiable path to national-scale qualification? The intended contribution is not a new consensus algorithm or a universal monetary-policy number. It is a composition and evidence methodology that connects each architecture claim to a formal, empirical, operational or institutional test.

- A sovereign-boundary architecture in which no foreign ledger or interlink can mint, burn or directly mutate another currency.
- A cross-border evidence protocol that binds payment intent, unequal-currency FX quote, bilateral compliance authorizations and final prepare receipts to explicit terminal certificates.
- A privacy evaluation with learned attackers, distribution shift and large ranking populations, plus a separately scoped person-level differential-privacy service for public statistics.
- Leakage-controlled AML/CFT experiments across independent graph families, calibrated sanctions-screening policy frontiers and prospective effectiveness outcomes.
- A partially identified, robust and adaptive policy-limit method that treats stability, utility, inclusion and distribution as constraints rather than assuming one transferable optimum.
- A four-stage national qualification method: analytical lower bounds, trace-driven queue simulation, physical multi-region testing and an independently observed pilot.

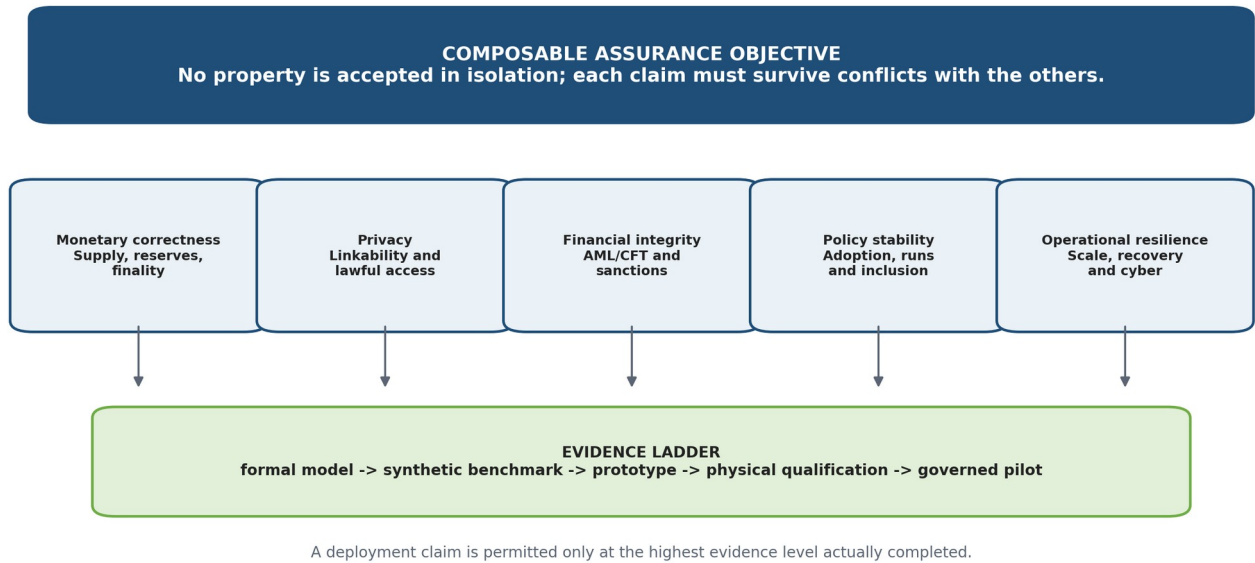


Figure 1. Composable assurance and the evidence ladder used throughout the paper.

2. Related work and novelty boundary

Project Icebreaker demonstrated cross-border retail settlement through two domestic CBDC legs funded by FX providers (BIS Innovation Hub 2023b). Project Mandala developed rules and proof engines for compliance-by-design (BIS Innovation Hub 2024). Project Agorá demonstrated atomic multi-currency wholesale settlement on a shared programmable platform and, in July 2026, completed 17 controlled real-value scenarios involving 28 financial institutions and central banks and approximately CHF 800,000; the BIS describes this as prototype testing rather than a finished product (BIS Innovation Hub and Institute of International Finance 2026). Project FuSSE reported 10,000 transactions per second in a controlled modular proof of concept while explicitly identifying scaling and operational challenges (BIS Innovation Hub 2026). These projects provide essential components, but they use different market structures and evidence scopes. Academic and policy work separately addresses atomic cross-chain CBDC settlement, TEE-based interoperability across independent semi-centralized ledgers, five-element cross-border design choices, functional CBDC architecture requirements, and privacy/AML regulation-by-design (Lee et al. 2021; Homoliak et al. 2023; Reslow, Soderberg, and Tsuda 2024; BIS Consultative Group on Innovation and the Digital Economy 2023; Pocher and Veneris 2022). A standards-based composable CBDC implementation already links token and contract networks (Bharathan and Pillai 2022); later work extends offline privacy/compliance classification (Michalopoulos et al. 2025), architecture-agnostic cross-system checkpoints (Mullins et al. 2025), and formal-method arguments for CBDC operational resilience (Bernardo et al. 2025). Jin and Xia's CEV framework already recommends CBDC consensus and operating architectures and validates proposals through experiments and formal analysis (Jin and Xia 2022). The claimed difference here is therefore not priority for any isolated component or for CBDC evaluation and verification as a broad category. It is the narrower joint claim-to-evidence qualification method that connects settlement safety, privacy leakage and differential privacy, AML and sanctions evaluation, adaptive holding limits, and falsifiable scale gates under one explicit maturity boundary.

The transaction-decision service builds on atomic-commit and fault-tolerant consensus literature rather than claiming a new consensus primitive (Gray and Lamport 2006; Castro and Liskov 1999; Yin et al. 2019). Privacy design follows the institutional and technical separation emphasized by NIST, IMF, Tourbillon and W3C work (NIST 2025; BIS Innovation Hub 2023d; W3C 2026; Murphy et al. 2024). Financial-integrity evaluation incorporates the network perspectives explored by Aurora and Hertha (BIS Innovation Hub 2023a; BIS Innovation Hub 2025) while distinguishing predictive performance from the effectiveness outcomes required by FATF (Financial Action Task Force 2026; Financial Action Task Force 2020; Financial Action Task Force 2023; Kao et al. 2025). Monetary-policy analysis is informed by BIS and ECB work on gradual and stress-driven bank disintermediation (Bidder, Jackson, and Rottner 2025; European Central Bank 2025b; European Central Bank 2025a). Operational qualification uses TIPS, Pix and FuSSE as workload references without treating any one reference as a universal CBDC demand forecast (BIS Innovation Hub 2026; European Central Bank 2024; Banco Central do Brasil 2025).

Table 1. Principal novelty boundaries.

Prior stream	Established contribution	Boundary addressed in this paper
Icebreaker (BIS Innovation Hub 2023b)	Two domestic cross-border legs and FX-provider competition	Evidence-bound prepare and terminal certificates; explicit recovery and legal evidence
Mandala (BIS Innovation Hub 2024)	Rules and cryptographic compliance proofs	Composition with monetary accounting, privacy, AML analytics and settlement finality
Agorá (BIS Innovation Hub and Institute of International Finance 2026)	Atomic multi-currency wholesale settlement on a shared platform	Independent-ledger isolation and jurisdiction-specific policy control
FuSSE (BIS Innovation Hub 2026)	10,000 TPS controlled modular settlement proof of concept	Qualification method that separates analytical, simulated, physical and pilot evidence

Prior stream	Established contribution	Boundary addressed in this paper
Paxos Commit / BFT (Gray and Lamport 2006; Castro and Liskov 1999; Yin et al. 2019)	Fault-tolerant decision and consensus foundations	CBDC evidence objects, sovereign boundaries and legal mapping
BIS/ECB policy analysis (Bidder, Jackson, and Rottner 2025; European Central Bank 2025b; European Central Bank 2025a)	Structural and bank-level CBDC exposure analysis	Partially identified, adaptive rule rather than universal transfer of one estimate

Scoped novelty. The contribution is the explicit composition of monetary accounting, privacy, financial integrity, independent-ledger finality, adaptive policy and evidence-gated scale qualification. Individual primitives are not claimed as novel.

3. System model, requirements and monetary invariants

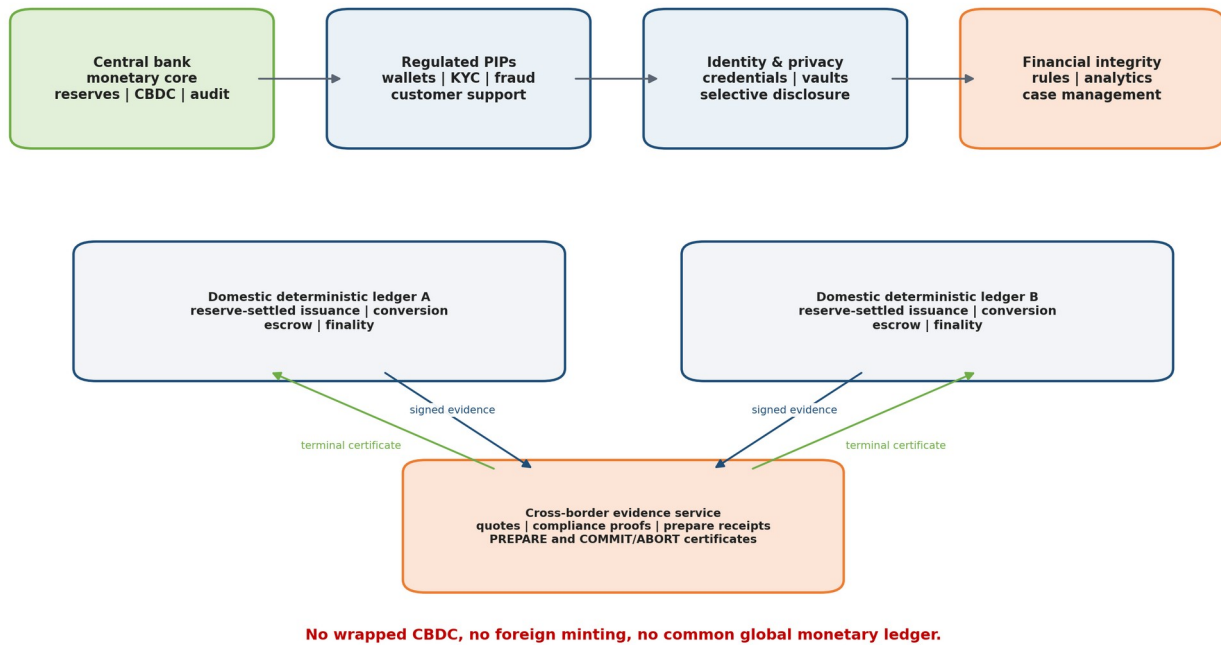


Figure 2. Technology-neutral architecture and the sovereign boundary.

Each domestic system contains a central-bank monetary core, a reserve/RTGS adapter, regulated payment-interface providers (PIPs), identity and credential services, financial-integrity services, an offline-risk module and operational-security controls. The implementation may use a high-assurance replicated database or a permissioned deterministic ledger. Governance remains centralized in law and monetary authority even when technical operation is distributed for resilience.

Table 2. Core design requirements.

Requirement	Operational interpretation
Monetary singleness	CBDC is redeemable at par and recorded as a direct central-bank liability.
Supply conservation	Issuance, redemption, escrow and offline allocation reconcile to unique sources and immutable receipts.
Two-tier accountability	PIPs retain customer-facing identity, support, fraud and AML responsibilities.
Purpose limitation	Each actor receives only the data required for its legal and operational role.
Deterministic finality	Terminal monetary state is technically and legally recognized; correction uses a new compensating transaction.
Failure containment	A foreign, PIP or analytics failure cannot create money or corrupt unrelated domestic balances.
Evidence before scale	Scope expands only after formal, privacy, AML, sanctions, economic, performance and legal gates pass.

Wholesale issuance converts reserve liabilities into CBDC inventory; retail conversion exchanges a commercial-bank deposit for existing CBDC inventory. These operations must not be conflated. Let R denote reserve liabilities, C issued CBDC, N notes, I_b the CBDC inventory of bank b , W_u a customer wallet and E settlement escrow.

$$\Delta C = -\Delta R \quad (1)$$

$$R + C + N = \text{constant (for a pure liability-composition change)} \quad (2)$$

$$C = \sum_b I_b + \sum_u W_u + E + O + A \quad (3)$$

Here O is risk-bounded offline value and A contains explicitly defined system accounts. Every increase in C references one final reserve or authorized accounting event. Replays are idempotent, balances cannot become negative, and finalized history is not rolled back.

4. Evidence-bound cross-border settlement

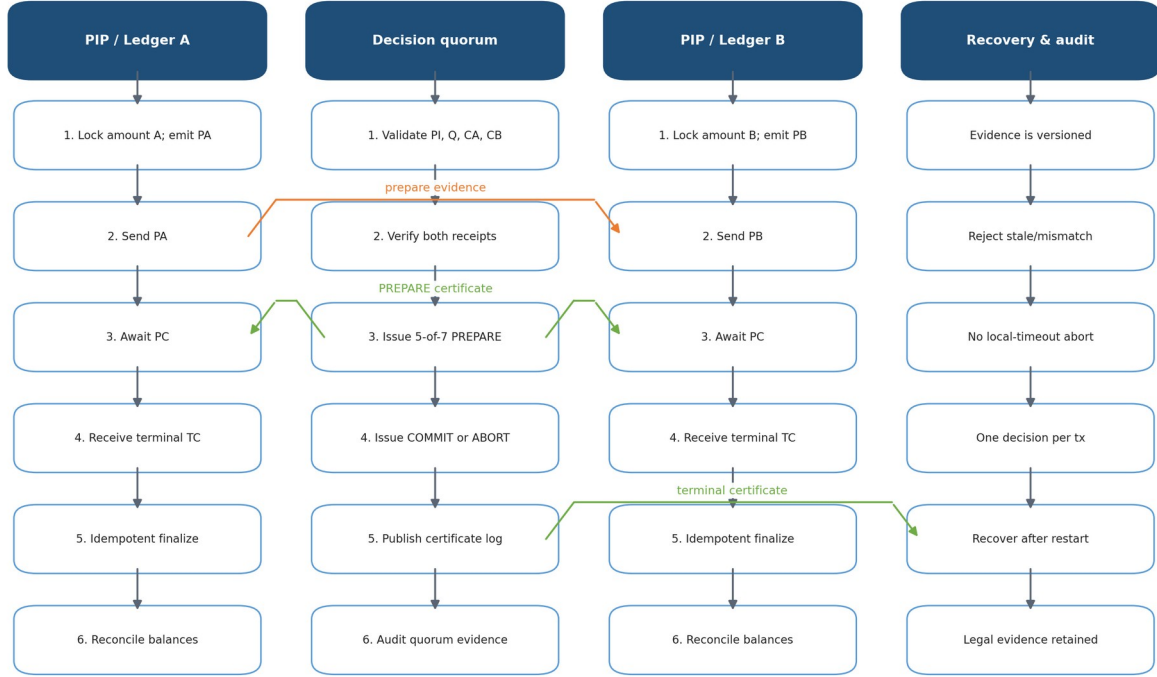


Figure 3. Cross-border sequence with explicit PREPARE and terminal decision certificates.

Currency A remains on ledger A and currency B remains on ledger B. A licensed FX provider is prefunded in both systems. The payer transfers A domestically to the provider; the provider transfers B domestically to the payee. The interlink coordinates signed evidence but never custodies or represents either CBDC.

Table 3. Cross-border protocol objects.

Object	Mandatory binding
Payment intent PI	payer/payee routing, currencies, maximum debit, minimum credit, corridor, expiry and unique transaction ID
FX quote Q	unequal amounts, rate, fee, provider, validity and signature
Compliance CA / CB	jurisdiction, policy and list versions, payment digest, amount and expiry
Prepare PA / PB	final domestic escrow state, exact currency/amount, source, destination and finality receipt
Prepare certificate PC	5-of-7 decision-node votes over the identical evidence digest
Terminal certificate TC	5-of-7 COMMIT or ABORT votes; honest nodes durably record one terminal decision

For $n=7$ signers and $q=5$, any two quorums intersect in at least $2q-n=3$ nodes. If at most $f=2$ keys equivocate and every honest signer durably signs at most one terminal decision for a transaction, conflicting valid terminal certificates require at least one honest signer to violate its lock.

$$|Q_1 \text{ intersection } Q_2| \geq 2q - n = 3 > f = 2 \quad (4)$$

Table 4. Cross-border properties and proof boundary.

Property	Argument and boundary
No local-timeout abort	A domestic ledger releases escrow only on a valid ABORT certificate; silence or local

Property	Argument and boundary
	time is insufficient.
No committed-versus-aborted finality	Both ledgers verify a terminal certificate bound to the same digest before finalization.
Compliance atomicity	PI, Q, CA, CB, PA and PB are included in the certificate digest.
Replay resistance	Unique IDs, expiry, digest binding and idempotent state transitions reject duplicate economic effects.
Sovereign isolation	The interlink cannot mint, burn, freeze or directly transfer either domestic currency.
Proof boundary	Safety assumes signature security, durable honest locks, correct verification and no more than two equivocating keys; indefinite denial of service and dynamic membership are outside the model.

The executable abstract model reaches 2,013 states through 19,480 transitions and finds no encoded conflicting-certificate or split-finality state. The eleven-service laboratory prototype completes 84 unequal-currency transfers and rejects stale compliance, mismatched quotes and conflicting certificates. These results establish bounded model and laboratory evidence, not certified multi-region or Byzantine deployment.

5. Privacy architecture and registered differential privacy

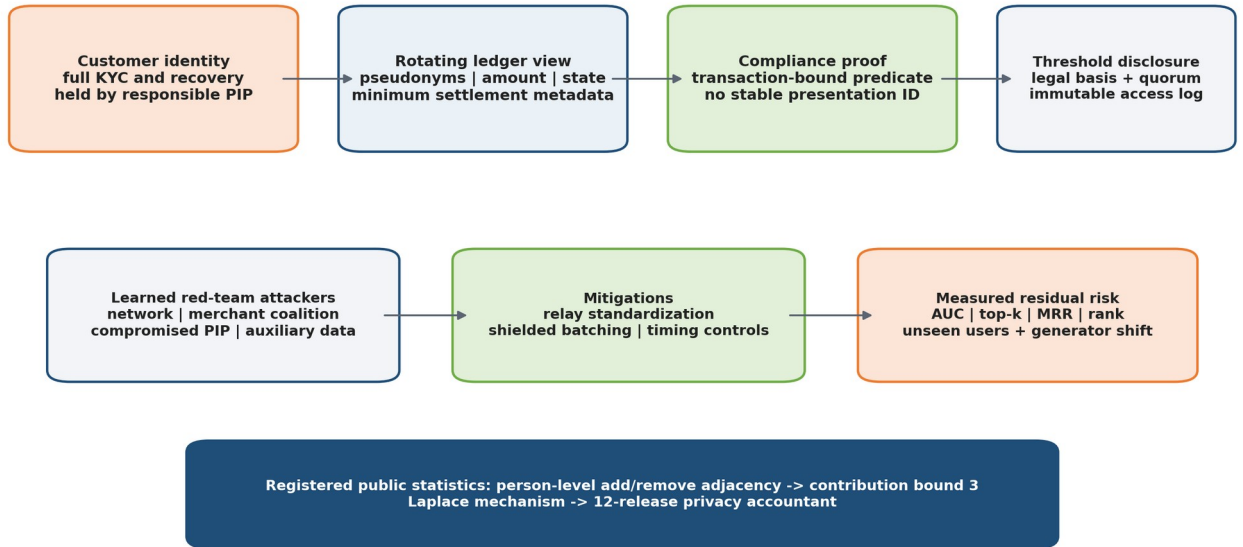


Figure 4. Institutional separation, learned privacy attacks and registered public-statistics releases.

5.1 Formal objective and attacker model

The privacy objective is data minimization and bounded linkability, not universal anonymity. Full identity remains with the responsible PIP. The core ledger receives rotating references and minimum settlement state. Compliance presentations are bound to a transaction and must not expose stable presentation identifiers. Lawful re-identification requires a documented legal basis, an authorization quorum and an immutable access record.

Table 5. Privacy attacker model.

Attacker	Observable information	Primary inference risk
Ledger operator	rotating references, amount, time, state and counterparty class	behavioral linkage across rotations
Network observer	endpoint class, transport timing, cadence and size bands	traffic correlation
Merchant coalition	repeat merchant visits, time and spending pattern	cross-merchant profiling
Compromised PIP	identity mapping and local history plus auxiliary data	direct re-identification and cross-system linkage

Let b be a hidden bit selecting whether two observations originate from the same person. An attacker A receives only the declared leakage L and returns a guess. The linkability advantage is:

$$\text{Adv_link}(A) = |\Pr[A(L_0, L_1) = b] - 1/2| \quad (5)$$

The formal objective requires negligible advantage for protected contexts under the chosen cryptographic and traffic assumptions. The empirical experiment deliberately relaxes those ideal assumptions and measures how much metadata remains exploitable.

5.2 Learned red-team evaluation

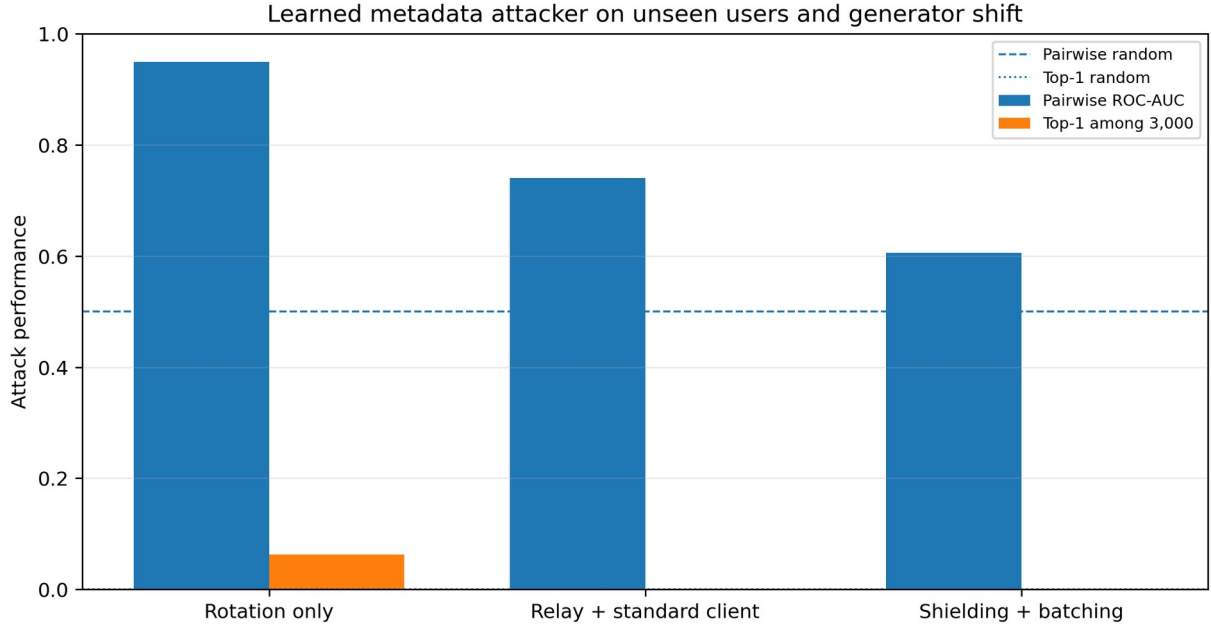


Figure 5. Learned pairwise-attack ROC-AUC under unseen-user and independently generated evaluation regimes.

Table 6. Learned network-observer linkage results.

Privacy profile	Evaluation split	Network-observer AUC	Seed range
rotation only	independent generator	0.950	0.949-0.951
rotation only	unseen users	0.992	0.991-0.992
relay standardized	independent generator	0.741	0.736-0.745
relay standardized	unseen users	0.926	0.917-0.935
shielded batched	independent generator	0.605	0.595-0.615
shielded batched	unseen users	0.830	0.813-0.846

The attack uses traffic timing, endpoint classes, merchant repetition, wallet implementation fingerprints and behavioral summaries. The independent-generator split changes the calendar/payroll regime, while the unseen-user split holds out identities but preserves the generator. Pseudonym rotation alone is highly linkable. Relay standardization reduces linkability, and shielding plus batching reduces it further, but unseen-user AUC remains materially above random. The experiment therefore rejects any claim that pseudonym rotation or batching alone establishes anonymity.

Table 7. Ranking attack under independently generated traffic.

Profile	Top-1 at 3,000	Top-10	MRR	Median rank
rotation only	0.062	0.188	0.102	133.5
relay standardized	0.000	0.000	0.006	697.5
shielded batched	0.000	0.000	0.002	1215.8

5.3 Registered public-statistics differential privacy

Differential privacy is used only for registered aggregate publications, not as a blanket description of the ledger or AML system. Two datasets are adjacent when one person and all of that person’s registered contributions are added or removed. A person contributes one count to at most three cells per monthly release, so the L1 sensitivity of the histogram is three. The release mechanism is:

$$M(D) = \max(0, f(D) + \text{Laplace}(\Delta_1 / \epsilon_t)), \quad \Delta_1 = 3 \quad (6)$$

$$\epsilon_{\text{total}} = \sum_{t=1}^{12} \epsilon_t = 4, \quad \epsilon_t = 1/3 \quad (7)$$

Non-negative clipping is post-processing and does not weaken the formal guarantee, but it creates bias in sparse cells and is therefore measured. Ten thousand neighboring-dataset checks confirm a maximum L1 change of three in the implementation.

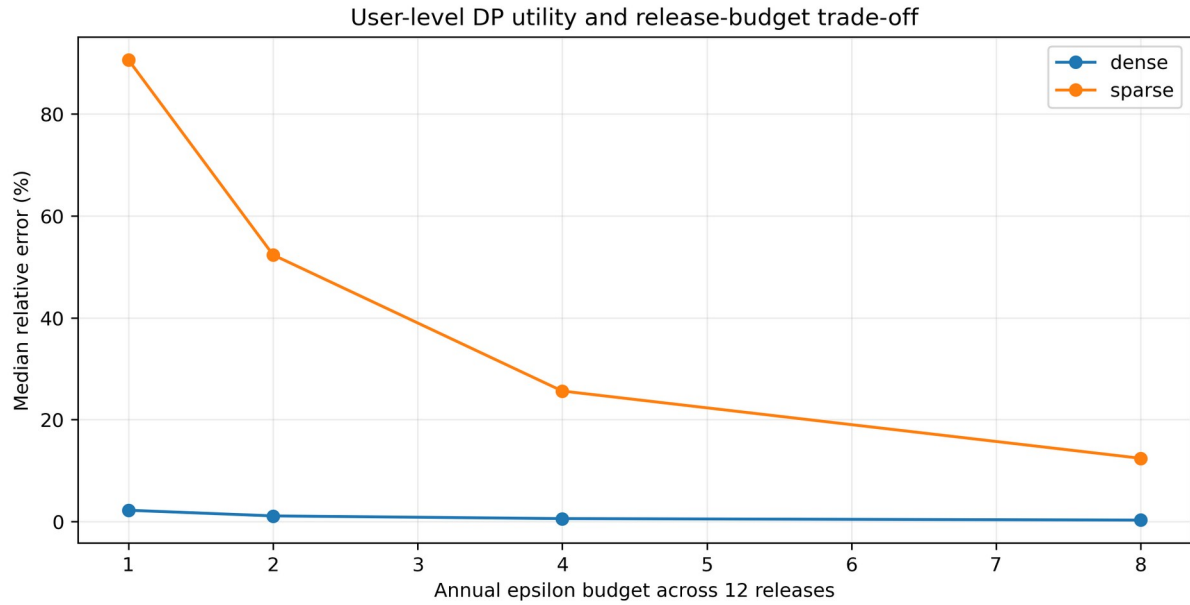


Figure 6. Utility and clipping effects for dense and sparse registered histograms.

Table 8. Differential-privacy utility by annual budget.

Annual epsilon	Dense MAE	Dense median relative error	Sparse MAE	Sparse median relative error	Sparse clipping bias
1	35.23	2.2%	25.86	90.6%	8.65
2	17.77	1.1%	15.63	52.3%	2.20
4	9.46	0.6%	8.68	25.6%	0.46
8	4.46	0.3%	4.19	12.4%	0.07

Guarantee boundary. The guarantee covers registered public histograms and their accountant. It does not cover raw data inside the trusted curator, identity maps, AML graphs, case files, arbitrary dashboards or unregistered revisions.

6. AML/CFT evaluation: predictive evidence and effectiveness gate

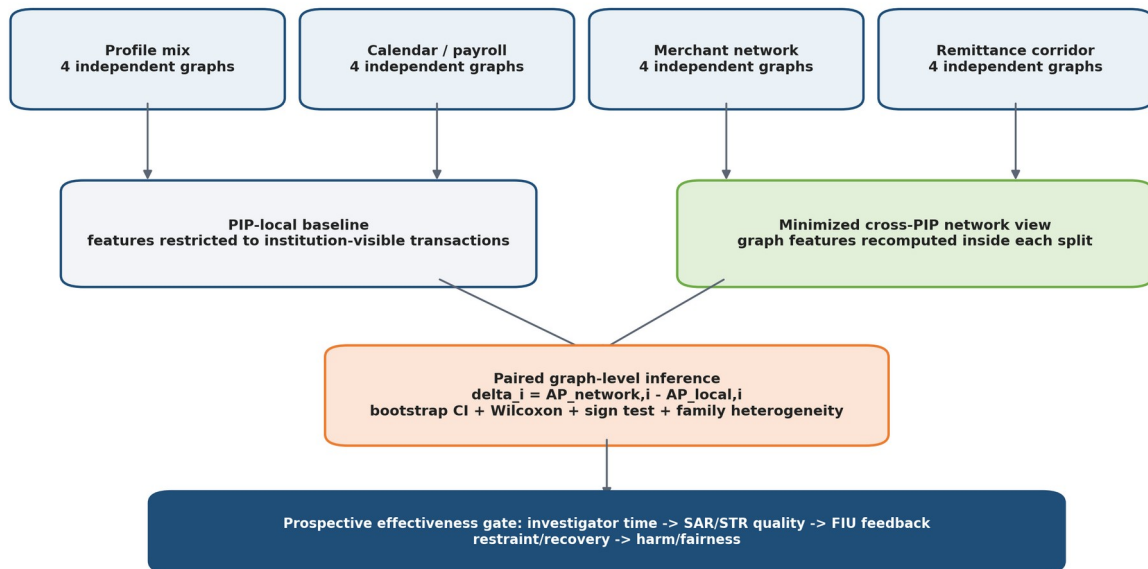


Figure 7. Leakage-controlled AML comparison and prospective effectiveness outcomes.

FATF evaluates effectiveness separately from technical compliance (Financial Action Task Force 2026). Accordingly, the experiment asks a narrower predictive question: do minimized cross-institution network features improve account-level ranking

relative to a genuinely PIP-local view across independent graph-generating regimes? It does not infer that a higher PR-AUC automatically improves investigations, suspicious-transaction reports or asset recovery.

Table 9. AML/CFT predictive-evaluation design.

Design element	Implementation
Independent experimental unit	Sixteen independently generated graphs; four graphs in each of four generator families
Families	profile mix, calendar/payroll, merchant network and remittance corridor
PIP-local baseline	features use only transactions visible to the institution
Network model	minimized cross-PIP graph features recomputed separately inside every train/test subgraph
Leakage control	account-group split and graph-local feature construction; labels and typology membership do not cross partitions
Robustness	family heterogeneity, adaptive dispersion/timing changes and workload-at-fixed-alert-budget
Inference	graph-level paired differences, bootstrap interval, Wilcoxon test and exact sign test

$$\text{delta}_i = \text{AP_network}_i - \text{AP_local}_i, \quad i = 1, \dots, 16 \quad (8)$$

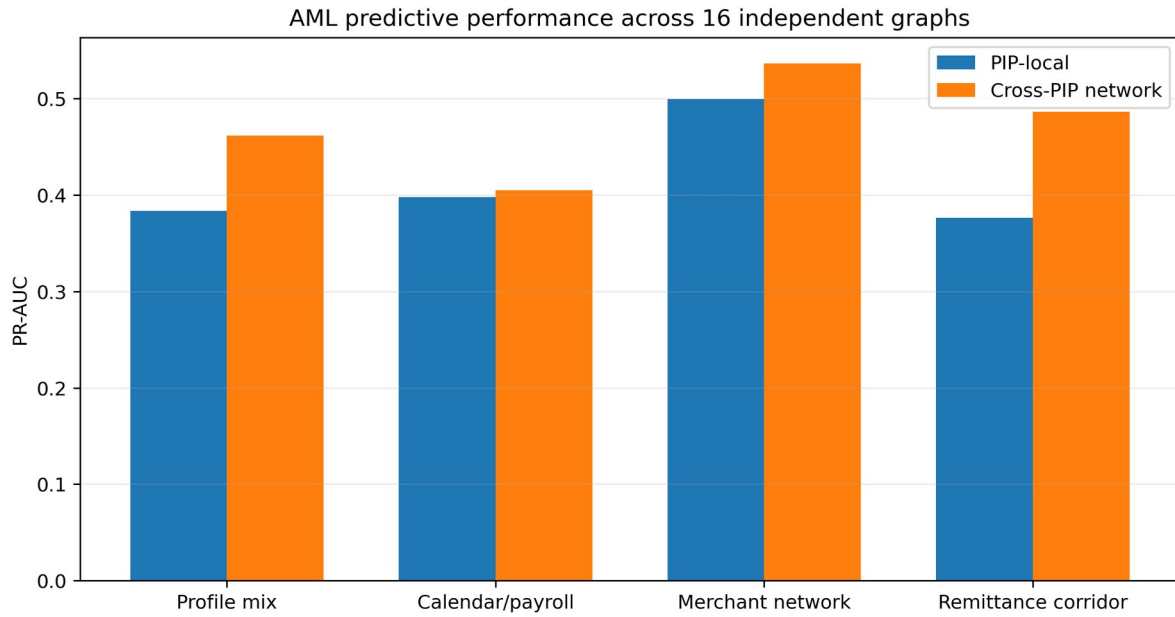


Figure 8. Mean PR-AUC difference by independent generator family.

Table 10. Family-specific AML effects.

Family	PIP-local mean AP	Network mean AP	Mean difference	SD of difference
calendar payroll	0.383	0.461	+0.078	0.020
merchant network	0.398	0.405	+0.007	0.086
profile mix	0.499	0.536	+0.037	0.092
remittance corridor	0.376	0.486	+0.110	0.067

Across all sixteen graphs, the mean paired difference is +0.058; the bootstrap 95% interval is 0.021 to 0.093, the paired Wilcoxon p-value is 0.0131, and the exact sign-test p-value is 0.0768. Twelve of sixteen graphs show a positive effect. The merchant-network family has a near-zero mean and high dispersion, while remittance-corridor graphs show the largest average gain. This heterogeneity is a central result: network analytics can help, but not uniformly.

Table 11. Predictive workload at fixed alert budgets.

Alert budget	Network recall	Network precision	PIP-local recall	PIP-local precision	Alerts / million
0.5%	0.014	0.625	0.012	0.562	5,000
1.0%	0.025	0.562	0.023	0.531	10,000
2.0%	0.048	0.531	0.046	0.516	20,000

Even where ranking improves, recall at small alert budgets remains low. Production evaluation must therefore be prospective and outcome-based: evidence sufficiency, investigator time, escalation, repeat alerts, SAR/STR quality, FIU feedback, restraint or recovery, network disruption, complaints and subgroup harm. Model outputs remain decision support and never automatic guilt determinations.

7. Sanctions evaluation as an operational decision system

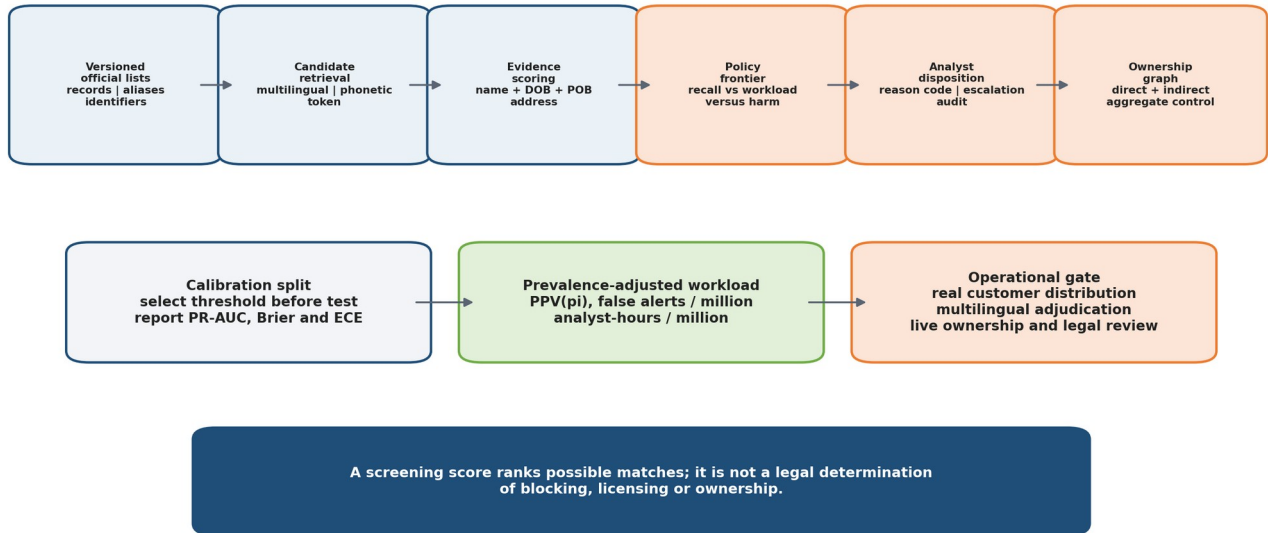


Figure 9. Sanctions screening from official-list ingestion to analyst and ownership decisions.

Sanctions screening is a staged decision process, not a string-similarity threshold. Official records and aliases are versioned; candidate retrieval uses multilingual, token and phonetic representations; an evidence model incorporates names, dates and places of birth, nationality, address and entity type; thresholds are selected on a calibration set; analysts decide disposition under the applicable programme and licensing rules; and ownership is evaluated separately, including direct and indirect aggregate control (U.S. Treasury, Office of Foreign Assets Control 2026b; U.S. Treasury, Office of Foreign Assets Control 2026c; U.S. Treasury, Office of Foreign Assets Control 2026a).

Table 12. Sanctions evaluation design.

Benchmark property	Implementation
Official input	19,178 OFAC records and 20,010 official aliases
Positive split	sanctioned entities disjoint across training, calibration and test
Perturbations	exact, official alias, OCR, token order, transliteration and sparse-attribute cases
Negative population	generated multilingual legitimate identities and hard near matches
Calibration	threshold selected before held-out test; Brier score and calibration error reported
Ownership	500 randomized monotonicity property tests for direct and indirect aggregate ownership
Boundary	no live customer population, independent multilingual analyst adjudication or live corporate registry

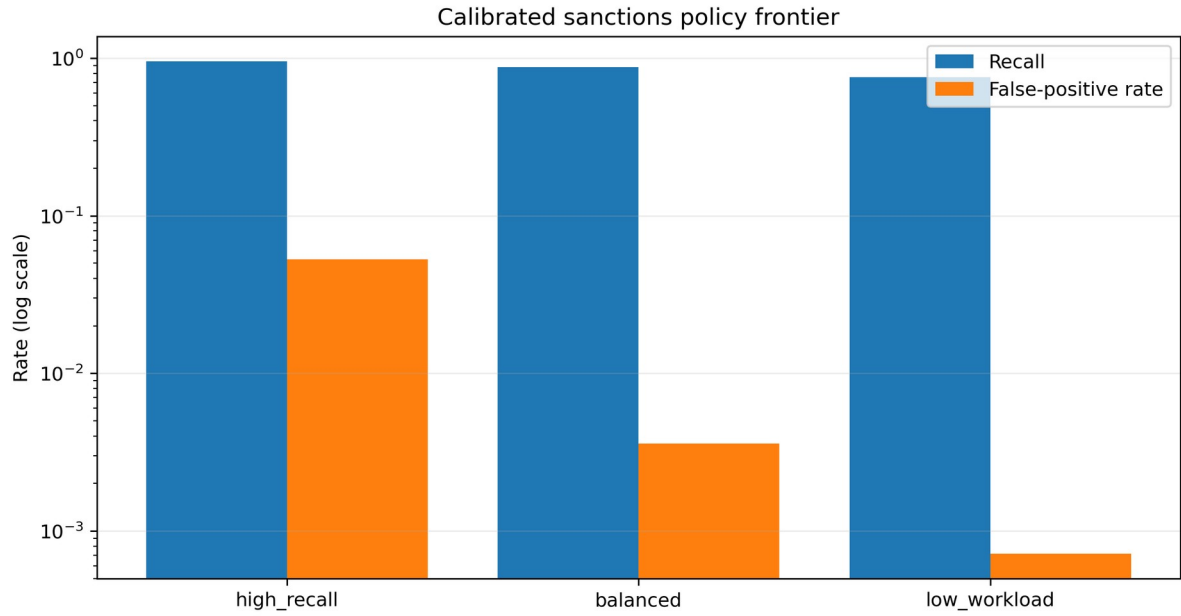


Figure 10. Calibrated sanctions policy frontier: recall and false-positive trade-off.

Table 13. Calibrated screening policies.

Policy	Threshold	Recall	False-positive rate	Benchmark precision
high recall	0.600	0.957	0.05286	0.808
balanced	0.731	0.874	0.00357	0.983
low workload	0.900	0.757	0.00071	0.996

$$PPV(\pi) = \text{Recall} \cdot \pi / [\text{Recall} \cdot \pi + \text{FPR} \cdot (1 - \pi)] \quad (9)$$

$$\text{Analyst hours} = (\text{true alerts} + \text{false alerts}) \cdot \text{minutes_per_alert} / 60 \quad (10)$$

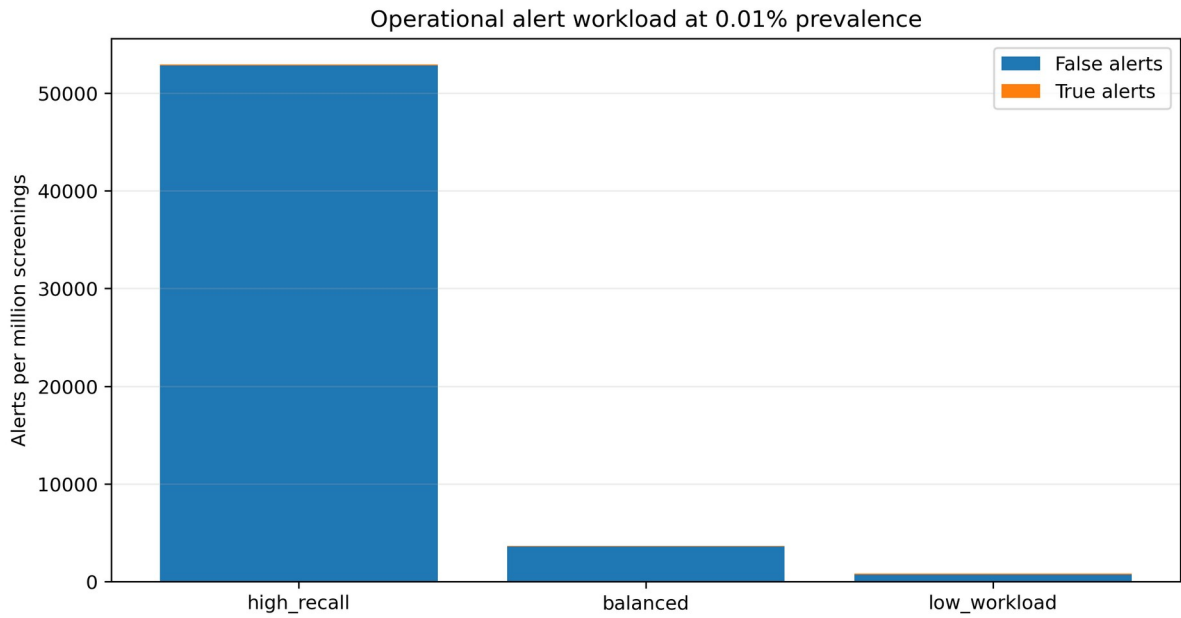


Figure 11. Prevalence-adjusted analyst workload across calibrated policies.

Table 14. Prevalence-adjusted workload illustration.

Policy at 0.01% prevalence	Operational PPV	True alerts / million	False alerts / million	Analyst hours / million
high recall	0.18%	95.7	52,852	6,177
balanced	2.39%	87.4	3,571	427
low workload	9.58%	75.7	714	92

Table 15. Recall by evidence and perturbation class.

Held-out perturbation class	Recall at balanced policy	Positive queries
exact	0.950	60
ocr	0.867	60
official alias	0.960	25
sparse attributes	0.617	60
token order	0.967	60
transliteration	0.933	60

The balanced policy obtains recall 0.874 and false-positive rate 0.00357 in the held-out benchmark. Benchmark precision is high because positives are deliberately overrepresented. At a 0.01% operational prevalence, the same rates imply only 2.39% positive predictive value and approximately 3,571 false alerts per million screenings. The low-workload policy reduces false alerts to approximately 714 per million but recall falls to 0.757. Sparse-attribute recall is 0.617. The real-world solution is therefore a risk-tiered frontier: high-recall policies for narrow high-risk cohorts, lower-workload policies for broad low-prevalence populations, analyst reason codes, workload caps, periodic calibration, and mandatory escalation when ownership or evidence is ambiguous.

8. Economic policy identification and adaptive guardrails

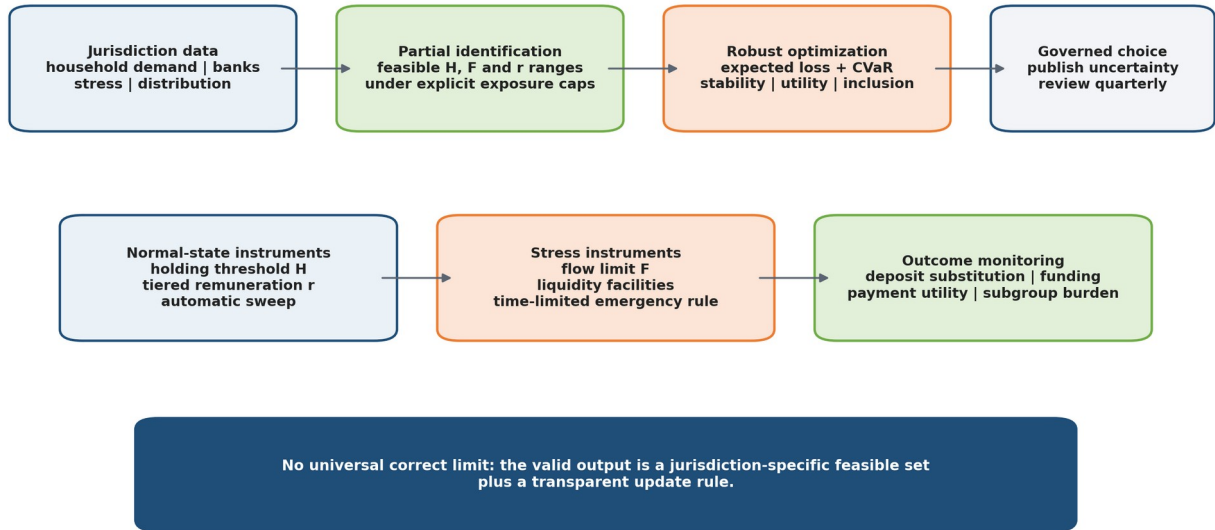


Figure 12. From jurisdictional data to a partially identified and adaptively governed policy set.

A universally correct CBDC holding limit cannot be inferred from public cross-country evidence. The relevant decision combines a holding threshold H , a system and user conversion-flow limit F , tiered remuneration r , automatic sweeps and liquidity facilities. The policy problem is partially identified because household demand, bank replacement funding, crisis behavior and distributional preferences vary across jurisdictions.

A transparent first constraint bounds gross CBDC exposure. Let D be eligible household deposits, N the eligible population, a the adoption share and τ the maximum accepted gross exposure share. Then:

$$H \leq \tau D / (N a) \quad (11)$$

Using the public euro-area deposit anchor $D = \text{EUR } 9.861$ trillion, $N = 300$ million, $a = 1$ and $\tau = 5\%$ yields an illustrative ceiling $H \leq \text{EUR } 1,643$. This is not a policy recommendation: lowering adoption to 50% doubles the arithmetic ceiling, and selecting 3%, 7.5% or 10% exposure tolerances changes it materially. The purpose is to expose the normative and empirical assumptions instead of hiding them in a single number.

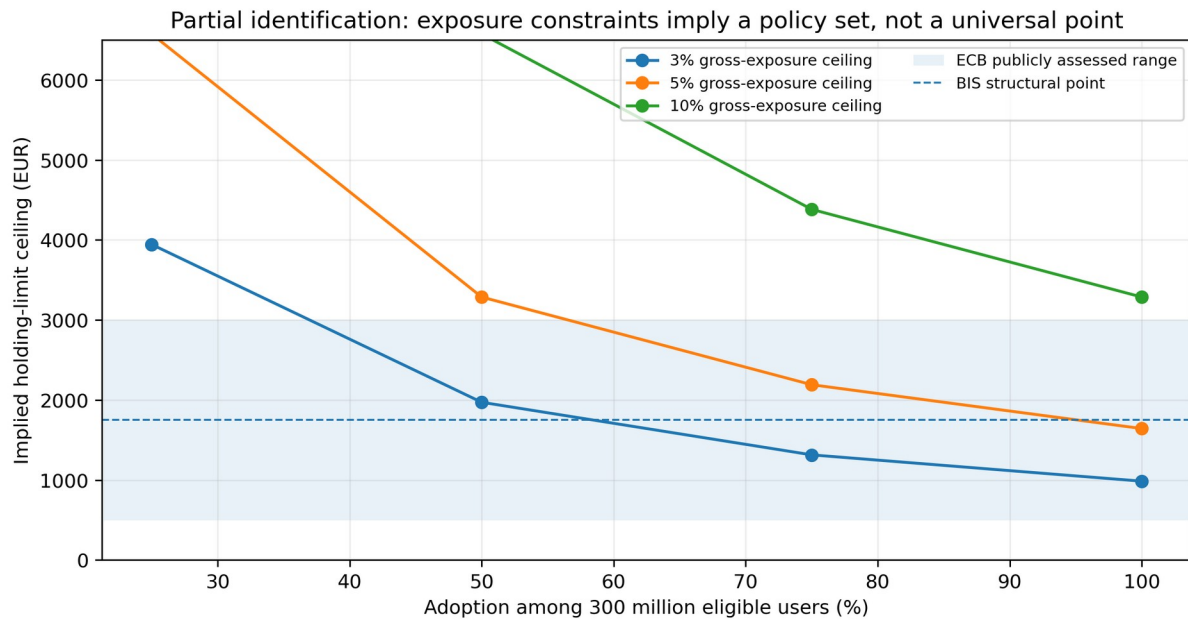


Figure 13. Illustrative holding-limit ceilings across exposure tolerances, eligible populations and adoption shares.

Table 16. Illustrative candidate limits under one exposure definition.

Holding limit	Gross exposure share (300m, full adoption)	Passes 5% cap	Passes 10% cap	Relation to BIS structural point
EUR 500	1.52%	True	True	+1250 EUR distance
EUR 1,000	3.04%	True	True	+750 EUR distance
EUR 1,500	4.56%	True	True	+250 EUR distance
EUR 1,750	5.32%	False	True	+0 EUR distance
EUR 2,000	6.08%	False	True	+250 EUR distance
EUR 2,500	7.61%	False	True	+750 EUR distance
EUR 3,000	9.13%	False	True	+1250 EUR distance

The policy authority should choose within the feasible set by minimizing expected and tail loss over an uncertainty set U . Let $\theta = (H, F, r)$ and L combine bank funding stress, payment-utility loss, exclusion and distributional burden:

$$\theta^* = \arg \min_{\theta \in \Theta} \max_{P \in U} \{ E_P[L(\theta, S)] + \lambda \text{CVaR}_{\alpha}[L(\theta, S)] \} \quad (12)$$

The optimization is subject to bank-level liquidity and funding constraints, ordinary-payment utility, accessibility, consumer-protection and legal constraints. A robust-regret or CVaR formulation does not create missing evidence; it makes uncertainty explicit. The parameter set must be updated on a predeclared schedule using survey data, pilot holdings, bank balance sheets, liquidity-facility use, deposit repricing, subgroup outcomes and stress simulations.

$$\text{Minimum binding days} = \text{ceil}(\text{desired conversion share} / \text{daily system flow share}) \quad (13)$$

Table 17. Mechanical duration imposed by system-wide flow limits (days).

Desired conversion share	0.25% daily limit	0.5% daily limit	1% daily limit	2% daily limit
2%	8	4	2	1
5%	20	10	5	3
9%	36	18	9	5

Policy conclusion. The valid scientific output is a jurisdiction-specific feasible set, uncertainty interval and transparent update rule - not one global holding-limit number.

9. National-scale qualification: from lower bounds to physical evidence

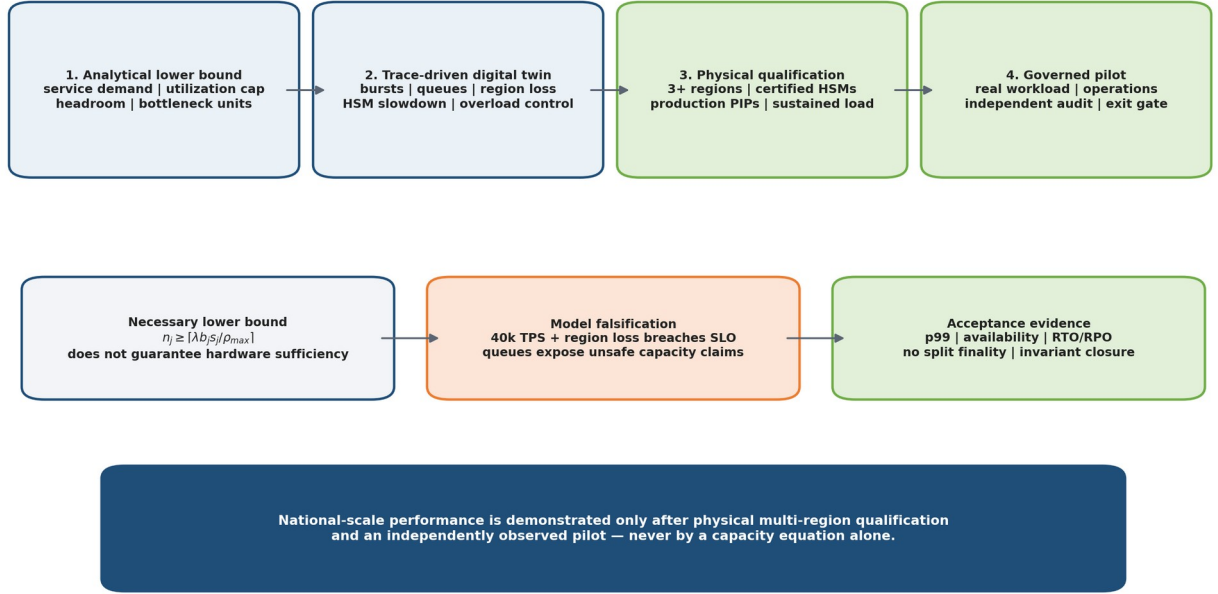


Figure 14. Four-stage national-scale qualification and the claim boundary.

A capacity equation cannot guarantee sufficient hardware because service times, queueing, replication, synchronization, HSM saturation, database contention, network loss and recovery interact nonlinearly. The paper therefore uses a hierarchy. Stage 1 derives a necessary analytical lower bound. Stage 2 runs a trace-driven tandem-queue digital twin with bursts and faults. Stage 3 executes the same predeclared workload on physical multi-region infrastructure with certified cryptographic hardware. Stage 4 verifies operations under a controlled external pilot. National-scale performance is demonstrated only after Stages 3 and 4 pass.

For stage j with qualification arrival rate λ , mean service demand s_j , headroom multiplier h and maximum planned utilization $\rho_{\max}$, a necessary lower bound on parallel units is:

$$n_j \geq \lceil \lambda h s_j / \rho_{\max} \rceil \quad (14)$$

This lower bound is intentionally not a guarantee. In the digital twin, each stage is represented as a finite-server queue with time-varying arrivals. Approximate stage utilization is:

$$\rho_j(t) = \lambda(t) s_j / n_j \quad (15)$$

The system passes only when the empirical latency and recovery distributions satisfy predeclared service objectives under normal, burst, region-loss, HSM-slowdown and overload-control scenarios. The model includes API/authentication, policy/sanctions, HSM verification, monetary transition and durable audit stages. Its service demands are declared scenario inputs, not measurements of a production system.

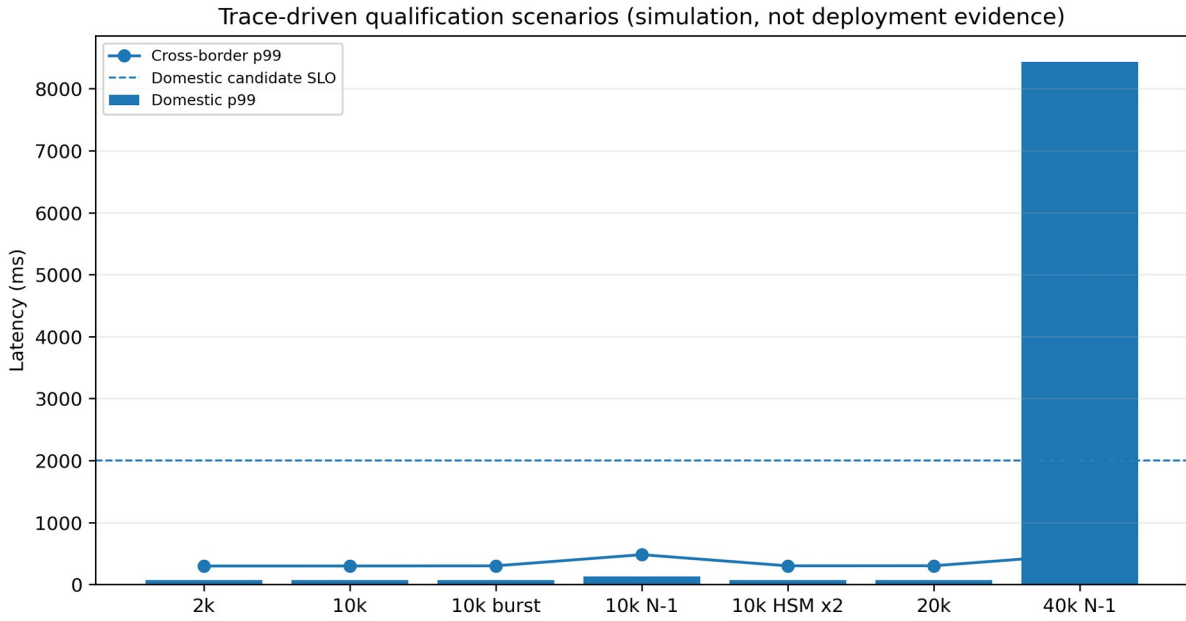


Figure 15. Trace-driven digital-twin p99 latency across workload and failure scenarios.

Table 18. Digital-twin scenario outcomes.

Scenario	Observed TPS	Domestic p99	Cross-border p99	Region loss	Domestic SLO
TIPS-reference steady	1,995	73.2 ms	298.0 ms	no	pass
FuSSE-reference steady	9,982	73.5 ms	298.6 ms	no	pass
FuSSE-reference burst	9,019	73.2 ms	300.5 ms	no	pass
10k with one region unavailable	9,721	128.6 ms	480.9 ms	yes	pass
10k with HSM service doubled	9,923	73.7 ms	300.7 ms	no	pass
20k qualification stress	18,616	74.0 ms	301.7 ms	no	pass
40k overload control with region loss	36,608	8,432.2 ms	479.8 ms	yes	FAIL

The declared 2,000-20,000 TPS scenarios satisfy the candidate two-second domestic p99 objective in the model. A 40,000 TPS overload with one region unavailable reaches p99 8.43 seconds and fails. This failure is valuable: it demonstrates that the model can falsify a capacity claim instead of being tuned to pass every scenario. Cross-border p99 remains below 0.5 seconds in the model because its modeled certificate path is not the congested bottleneck; physical qualification must verify whether this remains true with production consensus and cryptography.

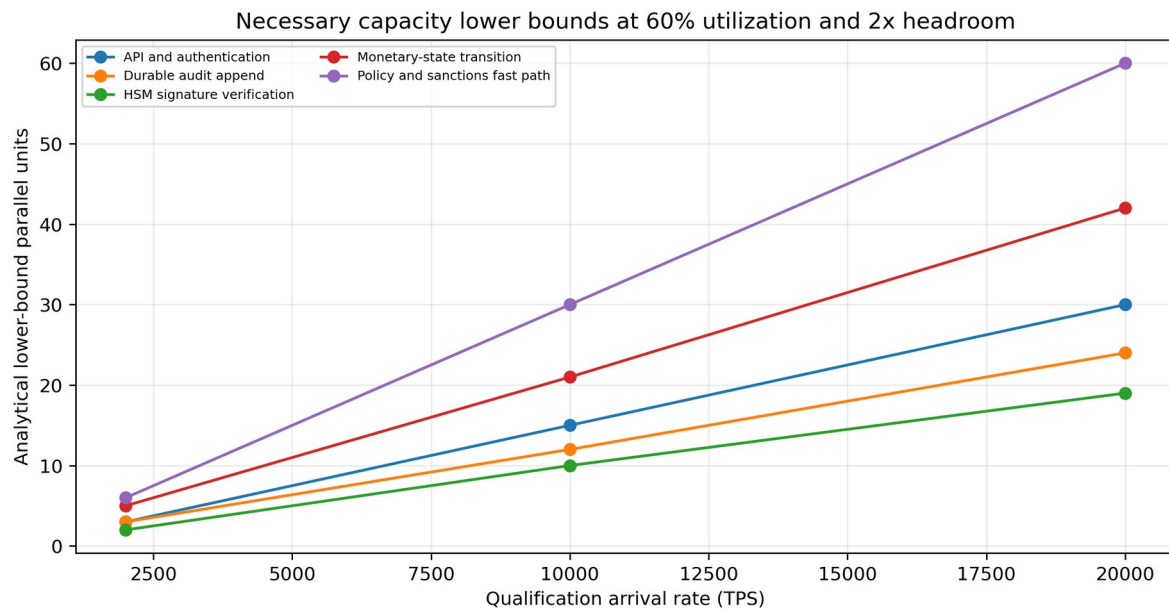


Figure 16. Analytical lower-bound units by stage and qualification workload.

Table 19. Necessary parallel-unit lower bounds (2x headroom, 60% planned utilization).

Qualification load	API/auth units	Policy/sanctions units	HSM units	Monetary units	Audit units
2,000 TPS	3	6	2	5	3
10,000 TPS	15	30	10	21	12
20,000 TPS	30	60	19	42	24

Table 20. Mandatory physical and pilot acceptance evidence.

Physical qualification dimension	Required evidence before a national claim
Topology	At least three independent regions/failure domains; measured inter-region distributions and operator map
Cryptography	Certified HSM throughput, threshold ceremonies, queue saturation, rotation, revocation and recovery
Workload	Open-loop and closed-loop PIP mix; realistic amount, identity, sanctions and reconciliation paths; sustained 2x observed peak
Faults	Region loss, partitions, stale rules, disk corruption, signer/PIP outages, network degradation and controlled overload
State safety	No monetary-invariant violation, duplicate economic effect, conflicting terminal state or unbounded escrow
Operations	RTO/RPO, reconciliation, fraud/support operations, change control, incident exercises and audit closure
External validation	Independent observer, reproducible load traces, public result summary and controlled pilot exit decision

National-scale status. The manuscript supplies an analytical bound, a reproducible digital-twin falsification test and a complete physical qualification protocol. It does not claim that national-scale performance has already been demonstrated.

10. Security, legal finality and governance

CBDC is critical national infrastructure. Security controls include threshold HSM keys, separation of duties, deterministic replication, immutable evidence, independent reconciliation, software bills of materials, reproducible builds, certificate transparency, privacy access logs and cryptographic agility (BIS Innovation Hub 2023c; NIST 2024a; NIST 2024b; Basel Committee on Banking Supervision 2021). General-purpose smart contracts do not receive issuance authority or unrestricted balance-write access. User-authorized conditional payments use audited templates and escrow primitives.

Table 21. Security controls and evidence.

Risk	Control and evidence
Issuance or decision-key compromise	threshold HSMs, amount/corridor limits, offline roots, witnessed ceremonies and rapid

Risk	Control and evidence
	revocation
Ledger corruption or equivocation	replication, state commitments, independent read replicas, reconciliation and restoration exercises
PIP compromise	signed customer intent, identity-vault segregation, least privilege and certificate revocation
Privacy insider abuse	purpose-based access, threshold disclosure, immutable logs and independent privacy authority
Model manipulation	data lineage, drift and evasion tests, reason codes, analyst review and model-risk governance
Supply-chain attack	SBOM, signed artifacts, reproducible build, independent review and staged release
Cryptographic transition	algorithm inventory, hybrid migration, re-signing plan and measured post-quantum overhead

Technical finality is insufficient without legal recognition. A corridor agreement must define the evidentiary status of PI, Q, CA/CB, PA/PB, PC and TC; quorum governance; liability for wrongful issuance or refusal; insolvency and FX-provider default; data transfer and sanctions duties; emergency suspension; dispute forum; and the domestic legal effect of finalization. A common protocol does not require identical laws, but each jurisdiction must map shared evidence to its own legal finality and supervisory framework (Cross-border Payments Interoperability and Extension Task Force 2026; ISO n.d.; CPMI-IOSCO 2012; European Union 2016).

11. Claims-to-evidence register, limitations and research agenda

Table 22. Claims-to-evidence register.

Claim	Evidence reported	Permitted conclusion	External gate
Cross-border safety	state exploration, quorum argument, negative tests and lab prototype	bounded safety under declared assumptions	formal refinement, certified HSM and physical multi-region fault testing
Privacy	learned synthetic attacks, unseen users, generator shift and large ranking sets	measured residual metadata linkability	governed real traces and independent red team
Differential privacy	explicit adjacency, sensitivity, utility and code-enforced accountant	formal guarantee for registered histograms	independent implementation and release-governance audit
AML/CFT	16 independent graphs, four families and graph-level inference	heterogeneous predictive improvement	prospective investigator, FIU, supervisory, disruption and harm outcomes
Sanctions	full official-list index, entity-disjoint test, calibration and prevalence workload	laboratory screening frontier	real customer distribution, multilingual adjudication and live ownership data
Economic policy	public anchors, partial identification and robust decision rule	conditional feasible set and update method	jurisdictional microdata, structural estimation and pilot evidence
National qualification	lower bound, digital twin, overload failure and acceptance protocol	falsifiable qualification methodology	physical regions, certified HSMs, sustained load and independent pilot

The principal limitations are deliberate and material. Privacy and AML experiments remain synthetic. The sanctions negative population is generated and does not reproduce every language, transliteration or customer-data quality problem. The economic illustration does not estimate behavioral parameters from microdata. The performance model uses declared service demands and cannot capture every implementation interaction. The cross-border fault model excludes indefinite denial of service, dynamic membership and post-compromise recovery. These limitations constrain the claims but make the research falsifiable.

The next empirical programme is therefore preregistered around six institutional studies: governed multi-PIP privacy red-teaming; prospective AML investigator and FIU outcomes; multilingual sanctions adjudication with real customer distributions; bank- and household-level economic estimation; physical multi-region/HSM performance qualification; and independent legal, cryptographic and operational audit.

12. Conclusion

A credible CBDC publication must distinguish architecture from evidence and evidence from deployment. The proposed system preserves monetary accounting and sovereign ledgers, binds cross-border finality to explicit evidence, quantifies rather than assumes privacy, restricts differential privacy to a formally specified release service, evaluates AML and sanctions under realistic workload boundaries, and treats the economic policy limit as partially identified and adaptive. For scale, the manuscript replaces a misleading hardware guarantee with a necessary lower bound, a trace-driven falsification model, physical acceptance criteria and an independently observed pilot gate. This combination provides a clean, reproducible and real-world research pathway while refusing to label laboratory results as national certification.

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Declarations of Interest

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